

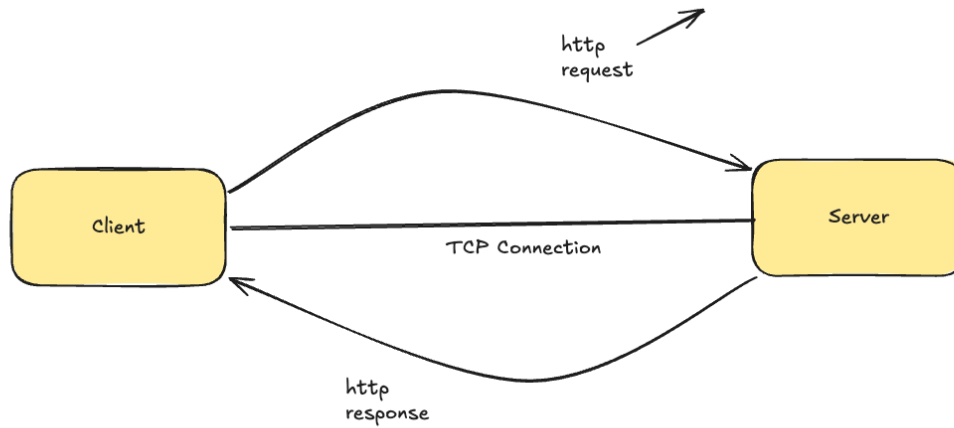
HTTP - Hyper text transfer protocol

1. It is an application layer network protocol.
2. It is a reliable protocol hence, dependent on TCP protocol in the transport layer.
3. It is a set of rules that generally govern how can you send hyper text over a network between two machines.
4. It will define how a machine will format, read and transmit hyper text from one machine to another.
5. Throughout years, http has developed to become a more general purpose application layer protocol and is now capable of not just transmitting hyper text but mostly any kind of data images, text, json etc.

primary communication mechanism between machines.

Note: How we will use HTTP can be diff for diff cases, but under the hood most of the comm is gonna be done by http.

What is hyper text ?
Hyper text is a text/document that contains links (hyperlinks) to other text or documents.



both the http req and res has multiple components and the protocol defines what should happen in those components.

 protocol://ipaddr:port

http (1.0)

- there was no support of persistent connections

http(1.1)

- persistent connection
- response can be streamed in parts

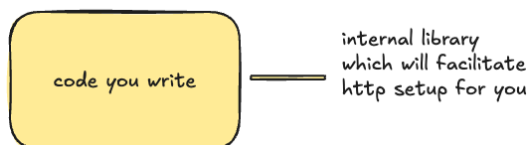
http (2)

- more perf improvements

http (3)

- uses quic instead of direct tcp

majority use cases

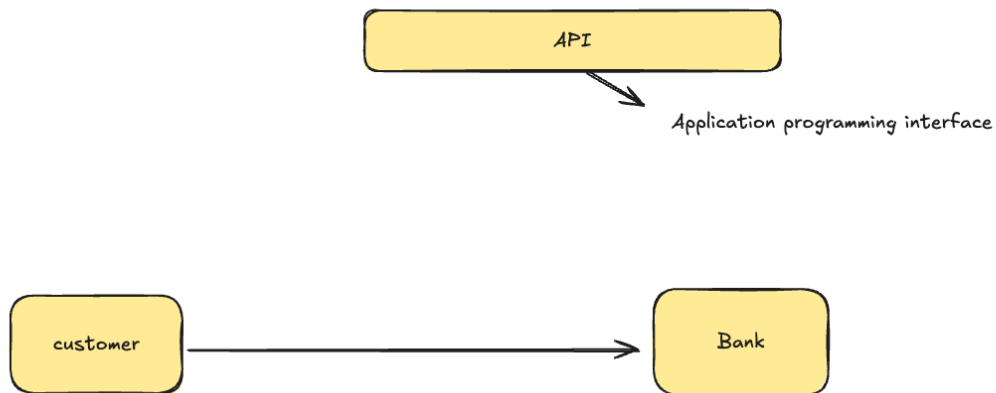


http request →

1. url -> url of the server to which we want to communicate
2. request headers -> extra meta data about the request, more like key value pairs
3. Request Method: its an enum that can be used to identify the nature of the request (GET, POST, PUT, PATCH)
4. Request body: it is a payload of information that we can send along with the request

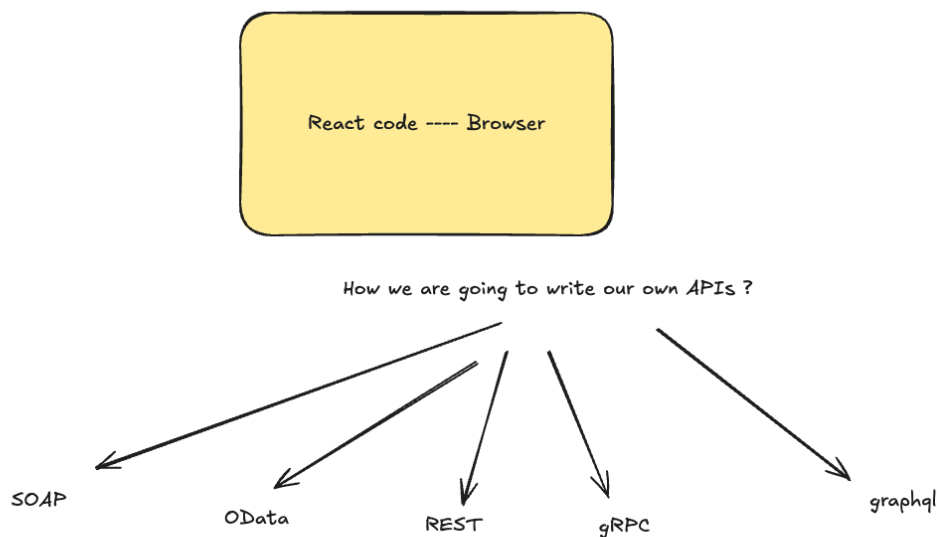
http response →

1. response body -> response of the server
2. response header
3. response code: numeric value, which can be used to identify the result of the request



What is an api ?

API is a mechanism with which the outer world will be able to interact with our software.



Top 3 most widely used ways

1. REST

2. gRPC

3. GraphQL